

Overview: NP Solution Limited deployed Hyperledger Fabric to create a decentralised platform for managing human resources and staffing services. The platform enables transparent and secure management of employee records, as well as payroll and recruitment processes.

Results: The use of Hyperledger Fabric has helped NP Solution Limited streamline HR processes, improve data security, and enhance trust among clients and employees.

4. IBM

Industry: Information technology and services

Overview: IBM leveraged Hyperledger Fabric to develop blockchain solutions for various industries, including finance, healthcare, and supply chain management. IBM's blockchain offerings enable clients to digitise and secure their business processes, improve transparency, and reduce operational costs.

Results: IBM's use of Hyperledger Fabric has enabled clients to achieve greater efficiency, security, and transparency in their business operations, leading to improved customer satisfaction and competitive advantage.

5. Walmart

Industry: Retail corporation

Overview: Walmart partnered with IBM to implement Hyperledger Fabric in its food traceability initiative. By digitising the supply chain, Walmart can track the journey of food products from farm to store shelves. This enables quick identification of the source of contamination in case of food safety issues, reducing food waste and enhancing consumer trust.

Results: The use of Hyperledger Fabric has significantly improved food safety standards at Walmart. The company can now quickly identify and remove contaminated products from its shelves, reducing the risk of foodborne illnesses and ensuring customer safety.

Creating a Hyperledger Fabric blockchain network: Get started

Step 1: Install prerequisites

Ensure you have the necessary software installed, such as Docker and Node.js.

```
# Install Docker (if not already installed)
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
```

```
# Install Node.js and npm (if not already installed)
sudo apt-get install nodejs npm
```

Step 2: Download fabric samples

Clone the Hyperledger Fabric samples repository from GitHub.

```
# Clone the Hyperledger Fabric samples repository from GitHub
git clone https://github.com/hyperledger/fabric-samples.git
```

Step 3: Generate the artifacts

Use the provided scripts to generate the cryptographic material and configuration artifacts.

```
# Navigate to the first-network directory within the fabric-
samples directory
cd fabric-samples/first-network
```

```
# Generate the cryptographic material and configuration
artifacts using the byfn.sh script
./byfn.sh generate
```

Step 4: Start the network

Use *Docker Compose* to start your network with the generated artifacts.

```
# Start the network using Docker Compose
./byfn.sh up
```

Step 5: Run your first application

Use the provided sample applications to interact with your Hyperledger Fabric network.

```
# Navigate to the chaincode-docker-devmode directory within
the fabric-samples directory
cd ../chaincode-docker-devmode
```

```
# Install the required Node.js dependencies
npm install
```

```
# Start the application using Docker Compose
docker-compose -f docker-compose-simple.yaml up
```

In conclusion, adopting Hyperledger Fabric is not just a choice; it's a strategic decision that can redefine the way businesses operate in the digital age. For developers, mastering this platform opens doors to innovative solutions that are secure, scalable, and tailored to modern challenges. For businesses, understanding Hyperledger Fabric is a pathway to efficiency, transparency, and a competitive edge in an increasingly digital world. **END** 🐧

By: Dhaval Gajjar

The author is the CTO of Textdrip, and the CEO of Pranshtech Solutions and WeTechnolabs Solutions. With a passion for technology and innovation, he explores the ever-evolving world of digital solutions, sharing insights and expertise to drive progress in the tech industry.